Equation discovery fails long before rate constants become unidentifiable in simulated adsorption columns

Abdulhamid Batayhi

Department of Mechatronics Engineering, Faculty of Mechanical Engineering,
Yıldız Technical University, Istanbul, Türkiye

ORCID 0009-0007-0374-6755

Abstract

Equation-discovery methods aim to learn governing laws from data, but how much more they need than fitting an assumed law is rarely measured. We measured it in a pre-registered computational study of adsorption columns. Over four decades of the Damköhler number Da (uptake rate relative to front passage), and across dispersion, sensor noise and isotherm error, we compared where the best of several sparse and symbolic regression methods stopped recovering the uptake rate law with where its rate constant stopped being identifiable. Wherever both limits fell within the tested range (37 conditions), discovery failed first, by 1.6 to 3.7 decades with 200 observations per run and still by 0.7 to 2.0 decades with 2,000. A 5 % isotherm error prevented discovery at every rate, while the rate constant stayed identifiable. Column data can calibrate an assumed rate law long after they can no longer reveal it.

Equation-discovery methods aim to learn a governing law itself from data, not only its coefficients: sparse regression over a library of candidate terms [1, 2] and its weak-form [3, 4], ensemble [5] and exact best-subset [6] variants, symbolic regression [7–10], bi-level identification [11] and neural networks embedded in the governing equations [12]. Discovering a law is a stricter task than estimating its coefficient, yet the two are rarely compared on the same data. Adsorption columns are a sharp test. A fixed-bed experiment reports concentrations and temperatures over time, most often at the outlet, and what an engineer needs from it is the uptake rate law, usually the linear driving force (LDF), $\partial q/\partial t = k\,(q^*(c,T) - q)$, with q the adsorbed loading, q^* its equilibrium value from the isotherm, c the concentration and T the temperature. For adsorption, one study has tried this from column data, training a neural uptake term on noisy outlet curves with the isotherm given exactly, at one rate and one noise level; for a Langmuir isotherm its readout held the two LDF terms and a constant the true law lacks, and for a Sips isotherm symbolic regression returned the exact LDF form [12]. Where such a result holds, and where it stops, is not known. The natural scale is the Damköhler number, $\mathrm{Da} = k\,t_{\mathrm{stoich}}$, the number of kinetic time constants in one passage of the adsorption front: at low Da uptake lags far behind equilibrium, at high Da it nearly keeps up.

Part of the answer is classical. Near local equilibrium the mass-transfer coefficient is poorly determined by column data [13]: finite rates then act only through an apparent dispersion [14, 15],

the LDF coefficient can be extracted wrongly from outlet breakthrough [16], and most of the output variability belongs to the isotherm [17, 18]. Yet lumped rate coefficients are routinely estimated from breakthrough curves [19, 20]. Batch studies warn that data near equilibrium cannot tell kinetic laws apart [21, 22].

Equation discovery has limits of its own. On a single transient that settles onto a slow manifold, as a breakthrough run does, the candidate terms become nearly collinear [23–25]. Noise induces phase transitions in what can be recovered [26, 27], some classes of dynamics cannot be discovered at all [28], and success maps against sampling and noise exist for biological data [29]. Discovery has been coupled to identifiability by keeping only identifiable models [30], identifiability has been extended from parameters to unknown functions [31], and kinetic rate laws have been discovered in reaction engineering [32]. How much harder discovery is, where, for which methods, and from which measurements has not, to our knowledge, been measured.

The question matters most where the law is not the textbook one. In the water-harvesting framework MOF-303 the water diffusion rate passes through a minimum at the steep step of the isotherm, where the LDF approximation is least accurate [33, 34], and a loading-dependent rate has been fitted by hand [35].

We asked, in a pre-registered computational study, for which regimes the adsorption rate law can be discovered from data a column experiment can produce, and whether discovery fails at the same Da as the practical identifiability of the rate constant [36], or earlier. The rate constant is practically identifiable when the data, fitted with the true law, confine it to within a factor of two. Scoring discovery requires knowing the true law, so the data were simulated: ground truth came from a verified non-isothermal column model [37] for a Langmuir sorbent and a MOF-303-like step isotherm. The design spanned four decades of Da, three levels of axial dispersion (Péclet numbers), four noise levels and four levels of isotherm error. The column was observed in three ways (Fig. 1): interior probes of concentration, uptake and temperature (O1), the same probes without uptake (O2), and the outlet alone (O3). Methods from each family above were run in the strongest configuration we could verify within declared cost limits, and identifiability was judged by profile likelihood on the same observations. Discovery succeeded when a method returned exactly the terms of the true law with the right signs and coefficient ratio. In each slice of the design (one isotherm, Péclet number, noise level and isotherm error) we located two boundaries, the Da at which discovery and identifiability fall to 50 % success, and called their distance in decades the gap. The design, thresholds and verdict words were frozen before any method ran on the grid; analyses chosen after the verdicts are labelled post hoc. In all 37 of the 96 slices in which both boundaries fell within the tested range, discovery failed first, 1.6 to 3.7 decades of Da before the rate constant stopped being identifiable, and it still failed first with ten times more observations.

Results

Discovery fails long before identifiability

We report the pre-registered comparison first, then what moves the boundaries, which methods discover the law, denser sampling, observation without probes of uptake, and a loading-dependent rate. Both isotherms passed the positive control (lowest Da, no noise or isotherm error): the law was recovered, and k identifiable, in 135 of 135 replicates for each. As Da grows, the driving force $q^* - q$ that distinguishes the law shrinks below the noise, while the curves still constrain k (Fig. 2).

The confirmatory test (H2b) compares, in each slice, the Da at which the best discovery method falls to 50 % success, $\mathrm{Da}^*_{\mathrm{disc}}$, with the Da at which k stops being identifiable in half the replicates, $\mathrm{Da}^*_{\mathrm{ident}}$. Of the 96 slices, 37 had both boundaries on the grid. In all 37 the verdict was "*discovery fails before identifiability*", so H2b's pre-registered overall word is the same (Fig. 3, Table 1; every slice in Extended Data Figs. 1 and 2). The gap was 1.6 to 3.7 decades of Da, with a median of 2.2 decades for the Langmuir sorbent (23 slices) and 2.5 for the step isotherm (14 slices); allowing for boundaries known only to lie between two levels, every gap is at least 1.35 decades (Methods). Discovery failed between Da = 0.11 and 13, while k stayed identifiable up to Da between 43 and 994. Uptake was still well resolved in time where discovery failed: kh, the rate constant times the interval between observations, was at most 0.26 at every discovery boundary and 0.85 to 20 at the identifiability boundaries (post hoc), although more samples did help discovery (below). The best discovery method was weak-form exact best-subset selection in 34 of the 37 slices and the weak-form ensemble in the other three.

Noise moved both boundaries, but not equally. With 0.5 % noise and an exact isotherm, discovery held to Da $\approx$ 4 to 7 and identifiability to Da $\approx$ 320 to 740; at 2 % noise discovery fell to Da $\approx$ 1, and at 5 % to Da $\approx$ 0.1 to 0.3, while identifiability stayed between Da $\approx$ 57 and 172. A small isotherm error hurt discovery more than noise did. At 0.5 % noise, adding a 0.5 % isotherm error moved $\mathrm{Da}^*_{\mathrm{disc}}$ from about 5 to about 0.1 to 0.2 (in the three slices that kept a boundary), while $\mathrm{Da}^*_{\mathrm{ident}}$ moved by less than 0.2 decade. With isotherm error, success was not monotone in noise: at 2 % isotherm error the best method reached at most 18 of 45 replicates without noise but up to 43 with noise: without noise, weak-form best-subset selection kept on average 3.6 to 4.0 terms (four is the maximum) at Da $\leq$ 10, absorbing the isotherm error, until noise lifted the residual above the selection criterion's floor (post hoc).

The 59 slices without a verdict lack a discovery boundary, not data (post hoc): in 40, discovery never reached 50 % at any Da, including all 24 slices with 5 % isotherm error; in 8, all without noise, it never fell below 50 %; in the other 11 the observed success crossed 50 % but the falling fit placed no crossing in range, most clearly for symbolic regression (next section).

Two checks support the verdict. First, with exact data the identifiability estimator recovered k to within 5 % at every Da up to 100 but was biased low at Da = 316 ($kh = 6.4$) and 1000 ($kh = 20$), with median estimates of 0.39 to 0.91 times k; without those two levels every slice still has a gap of at least 0.89 decade (post hoc; Extended Data Fig. 3). Second, when success also required the

fitted k within a factor of two of the truth (a pre-registered sensitivity analysis), all 37 verdicts held and two more slices gained the same verdict.

The two other confirmatory tests could not be decided

The second confirmatory test (H2a) asked whether one ratio, the realised driving force over the combined observation and isotherm error ($R = \delta/\varepsilon_{\mathrm{eff}}$), places the discovery boundary at the same R^* in every condition. Only one of twelve conditions crossed 50 % in R ($R^* = 123$, 95 % interval 113 to 135), so the test has no verdict (Extended Data Fig. 4).

The third (H2c) asked whether an errors-in-variables estimator moves the discovery boundary upward relative to the best strong-form comparator (18 slices at 2 % noise; Methods). No slice had a boundary for both, so the test has no verdict at any isotherm error and is neither supported nor falsified: in 16 of the slices the comparator succeeded in no replicate at any Da, and the errors-in-variables estimator reached at most 40 %.

Which methods discover the law, and where

Exact best-subset selection in the weak form was the most reliable method: its success crossed 50 % within the grid in 36 of the 96 slices, against 8 for weak-form sequential thresholding and 7 for the weak-form ensemble (Fig. 4c). No strong-form sparse method had a crossing in any slice. Symbolic regression, run only at Pe $\approx$ 94 with noise of 0.5 or 2 % and isotherm error of 0 or 2 %, behaved differently from all the sparse methods. It failed at the lowest rates, where weak-form sparse regression worked best, and succeeded at intermediate rates: up to Da $\approx$ 32 without isotherm error, and between Da = 1 and 10 with 2 % isotherm error, where no sparse method reached 50 % (Fig. 4a,b; Extended Data Fig. 5). The pre-registered boundary, a falling logistic curve, cannot describe such a window. If the boundary is instead taken as the last level with at least 50 % success (post hoc), H2b is unchanged: in the four such slices with a verdict, that level lay 1.0 to 1.4 decades below the identifiability boundary, and at least 0.5 decade below it if success held to the next level. Neither family covers the range in which k remains identifiable.

The neural-network pipeline of the closest prior work, run as a declared demonstration with one replicate per level, recovered the law in one of its nine fits (Da = 3.2). Implicit sparse regression without any isotherm (SINDy-PI), the pre-registered route to the law's form on an isothermal control, recovered it in none of 2,160 fits, even without noise: every fit kept six terms, one more than the implicit Langmuir–LDF relation. Slow-manifold discovery, which targets the isotherm, reached 50 % only with exact data, at Da = 1000 (Extended Data Fig. 5).

Where open discovery fails, a law can still be chosen among candidates. In a post hoc check, the LDF law, the quadratic driving force and the pseudo-second-order law were each fitted by the likelihood used for identifiability and the best fit chosen (Pe $\approx$ 94, noise 0.5 and 2 %, isotherm error 0 and 2 %). The LDF law was chosen in all 45 replicates at every Da $\leq$ 10 with an exact isotherm, and in at least 35 of 45 with 2 % isotherm error, where sparse regression failed at every rate (Fig. 4d). It did not reach the identifiability boundary: with an exact isotherm, the LDF law was chosen in

only 31 of 180 replicates at Da = 31.6 and in none from Da = 100 on; on noise-free data with up to 20 times more samples, the LDF law won in three of the four cells tested at Da $\geq$ 100 (post hoc; Supplementary Note 1).

Ten times more observations narrow the gap but do not close it

The boundaries above assume 200 observation times per run. With 400 (post hoc), every boundary moved to higher Da or stayed, and discovery still failed at least 1.2 decades first. With about 2,000, on the Pe $\approx$ 94 cells of both isotherms at 0.5 and 2 % noise (declared before it ran; Supplementary Note 1), discovery moved up by 0.7 to 1.9 decades and identifiability by 0.2 to 0.5 decade. Discovery still failed first in the three slices with both boundaries, by 0.7 to 2.0 decades, above the 0.25-decade threshold; in the fourth (step isotherm, 0.5 % noise) discovery reached Da $\approx$ 310 while k stayed identifiable beyond the grid.

Without probes of uptake, k stays identifiable but the law is not discovered

With concentration and temperature probes but no uptake (O2), uptake must first be reconstructed from the gas-phase mass balance. Discovery then failed everywhere: no method succeeded in more than 29 % of the replicates at any level of any slice. Identifiability, computed on the subgrid where every point of the profile is a full column solve (Pe $\approx$ 94, noise 0.5 and 2 %, isotherm error 0 and 2 %), held to Da $\approx$ 57 to 286 from probes without uptake, and to Da $\approx$ 38 to 172 from the outlet alone (O3). With three replicates per level, most of these boundaries are known only to lie between two levels. In the same slices, the last level at which any method discovered the law from probes of c, q and T, a far richer observation, lay at Da = 3 to 32 (post hoc). The outlet curve alone therefore still determined k at least as far as that level in every slice, and up to 1.5 decades beyond it.

A loading-dependent rate passes the test almost only at Da = 1

For the step isotherm the study also simulated a rate that falls to 20 % of its value across the isotherm step (L2), a form motivated by the measurements in MOF-303; this part is exploratory. By the pre-registered rule, the state dependence was discovered almost only at Da = 1. There the best method succeeded in 77 to 100 % of all 60 replicates at every noise level and isotherm error; at any other level it reached at most 58 %, and without noise it never succeeded at Da $\leq$ 0.32 (Extended Data Fig. 6). These are best values over methods and Péclet numbers, so they bound any single method from above. A likely reason lies in the test itself: the library cannot represent a rate falling sigmoidally across the step, so success follows which polynomial terms are selected (Methods).

Discussion

Column data can calibrate an assumed rate law long after they can no longer reveal it: the law could be discovered only over a narrow part of the range in which its rate constant could be measured, 1.6 to 3.7 decades of Da short of it. For practice this sets an order of operations in three steps (Fig. 5). First, where uptake is slow compared with the passage of the front (Da of order one or below) and the isotherm is known almost exactly, weak-form sparse regression can propose the law; with as little as 0.5 % noise and 0.5 % isotherm error, every discovery boundary fell to Da ≤ 0.23. Second, up to Da ≈ 10 the law can still be chosen among a few candidates by likelihood, even with 2 % isotherm error. Third, beyond Da ≈ 10 the law has to be assumed. The data still give its coefficient: from interior probes for one and a half to nearly four decades of Da beyond the discovery boundary, and, in the few conditions tested, at least as far from the outlet alone. A misplaced isotherm step is the exception (see the last paragraph).

Discovery is more fragile because the two tasks ask different things of the same data. Identifiability asks whether a known law with one free coefficient fits well only near the true value. Discovery asks the data to choose that law among many and to return the coefficients of q^*_{meas} and q in the ratio -1 within 10 %. An isotherm error enters discovery as an error in a regressor, which biases exactly that ratio. In the identifiability fit the same smooth error is absorbed into k, and the estimate stays within a factor of two of the truth. The response to a 0.5 % isotherm error, more than a decade for discovery and under 0.2 decade for identifiability, is consistent with this reading, though the study was not designed to separate the two. The asymmetry of the two success rules does not make the gap: dropping the ratio test, requiring the interval to lie within 10 % of k, or judging by the point estimate alone kept every verdict, with gaps of 0.89 to 3.75 decades (post hoc). A second effect is classical: as Da grows, uptake approaches equilibrium and the law's two terms become nearly collinear on the slow manifold [24, 25]. In a post hoc check, the smaller singular value of the standardised pair $\{q, q^*_{\mathrm{meas}}\}$ fell from about 0.75 of the larger at Da $= 0.1$ to about 0.001 at Da $= 1000$. Exact data still separate the two terms, which is why discovery never failed with exact data (no noise and no isotherm error), but noise and isotherm error soon exceed the shrinking direction that distinguishes them. A one-parameter fit needs no such direction: it uses the whole curve to place k. We expect the same asymmetry wherever a law is to be discovered alongside an imperfectly known constitutive input, such as an equation of state or a transport property, but this study tested only adsorption.

Real columns sit mostly beyond the discovery boundaries. Published data for water on MOF-303 [33] and carbon dioxide on zeolite 13X [38] or MIL-91(Ti) [39] place physisorbent packed beds between Da ≈ 100 and 23,000 (order-of-magnitude estimates; Supplementary Table 1). An amine bed for direct air capture lay at Da ≈ 8 to 10 [40], within reach of the highest discovery boundaries (Da ≈ 7 with noise, 13 without). With 200 observations per run the law must be chosen or assumed for physisorbent beds; with 2,000 and 0.5 % noise, discovery reached the lower end of their range.

The closest prior work [12] is one point against this map. From its published equations (pre-registered, computed after the verdicts), its columns ran at Da ≈ 8.8 (Langmuir) and 9.2 (Sips)

with 5 % outlet noise and an exact isotherm, above every discovery boundary we found with noise. Under our rule its Langmuir readout (the LDF terms plus a constant) fails, but its Sips readout, $0.198\,q^{*} - 0.200\,q$, succeeds. That is one realisation of an isothermal model with its own error model, and a replicated test at that point would show whether the difference lies in the model, the errors or chance.

Whether a loading-dependent law such as that of water in MOF-303 [33, 34] can be discovered remains open. Our library could not represent a sigmoidal $k(q)$, so that test measured which polynomial terms were selected rather than whether such a law is discoverable.

All data were simulated by one verified column model with the LDF law as the truth, so the boundaries hold for that model, the two isotherms and the error models used; no second physical system was tested. Real columns add errors in geometry, dispersion and heat transfer that were held exact here, and real isotherms are rarely known to 0.5 to 5 %; both would be expected to lower the discovery boundaries, as would a larger candidate library. Interior probes of uptake (O1) are a ceiling that experiments rarely reach. The identifiability criterion measures precision and accuracy, not calibration: the intervals contained the true k in only about 10 % of O1 rows (23 % on O3; exploratory; Extended Data Fig. 7), yet judged by the point estimate alone every verdict holds. Discovery gains most from denser sampling, so the boundaries at 200 observations per run understate what a densely sampled experiment can discover. Symbolic regression ran on a subgrid, the neural pipeline only as a demonstration, and a Kolmogorov–Arnold network only in an exploratory run, where it recovered the law in none of 216 fits; each limit was set by cost rules declared before the grid ran, on the one computer available, and the larger neural pipeline declared for a graphics processor was not run. Combining method families was not tested, nor were errors-in-variables fitting in the weak form or recent methods such as parallel symbolic enumeration [9] and bi-level identification [11]. Errors found in our own code were corrected and are reported beside the results they replaced (Methods).

Three directions follow. Discovery would gain most from better isotherms and denser sampling. In a post hoc test, shifting the step of the MOF-303-like isotherm by 1 % relative humidity kept discovery at or below 31 % at every rate, biased the fitted k by a factor of 1.5 to 1.8 at $\mathrm{Da} = 10$ with intervals too narrow to show it, and lowered the identifiability boundary to $\mathrm{Da} \approx 17$ to 57 (5.6 with a 2 % shift). An isotherm whose step is placed to better than 1 % relative humidity is worth more than quieter sensors. And the map should be tested on experimental breakthrough data with interior probes at moderate Da, where it predicts that discovery first becomes possible.

Methods

Ground truth

Ground truth came from the finite-difference reference solver of the companion study [37]: a one-dimensional, axially dispersed, non-isothermal model of a packed column, verified there against

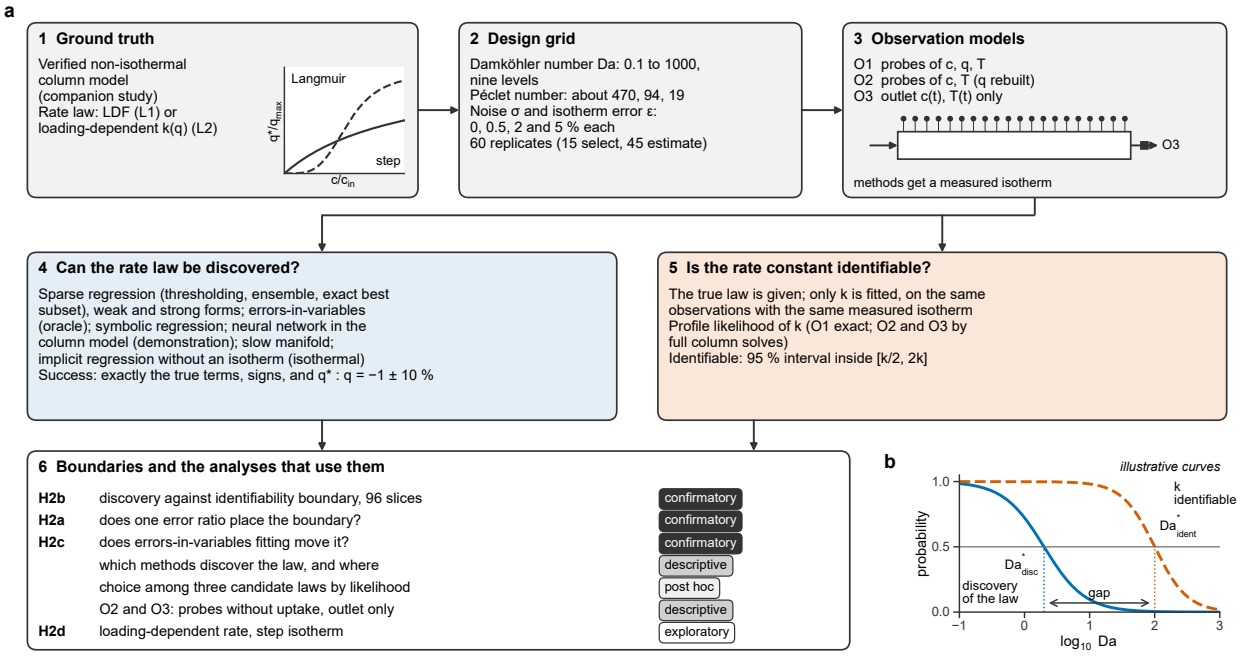


Figure 1: **Study design.** **a**, Breakthrough data are simulated by a verified column model for a Langmuir sorbent and a MOF-303-like step isotherm (box 1; isotherm shapes drawn from the model), over the design grid (box 2), and observed through interior probes with uptake (O1), probes without uptake (O2) or the outlet alone (O3) (box 3). The same observations, with the same measured isotherm, go to the discovery methods (box 4) and to a profile-likelihood test of k with the true law given (box 5). Box 6 lists the analyses and their evidence type. **b**, What the confirmatory test compares in each slice: the Da at which the best discovery method falls to 50 % success ($\mathrm{Da}^*_{\mathrm{disc}}$) and the Da at which k stops being identifiable in half the replicates ($\mathrm{Da}^*_{\mathrm{ident}}$); illustrative curves.

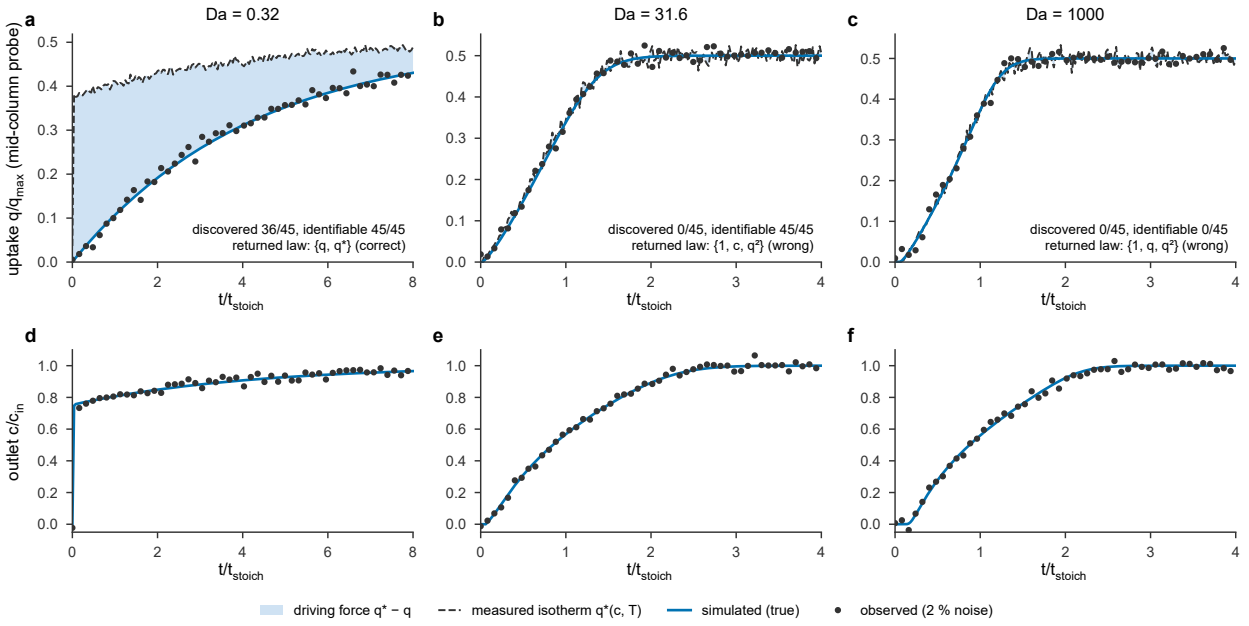


Figure 2: **What the data look like where discovery works, where only identifiability works, and where neither does.** Langmuir sorbent, Pe ≈ 94, 2 % noise and an exact isotherm, at Da $= 0.32$, 31.6 and 1000 (columns). **a**–**c**, A mid-column probe as the methods see it: observed uptake (points, every fourth time shown), simulated uptake (line) and the measured isotherm evaluated on the observed c and T (dashed); the shaded band is the driving force $q^* - q$, which shrinks below the noise as Da grows. Each panel gives the number of the 45 estimation replicates in which the law was discovered and k was identifiable, and the law the best method returned for one replicate. **d**–**f**, The outlet breakthrough curve of the same replicate. Time axes span each run's solved horizon (8 or 4 stoichiometric times).

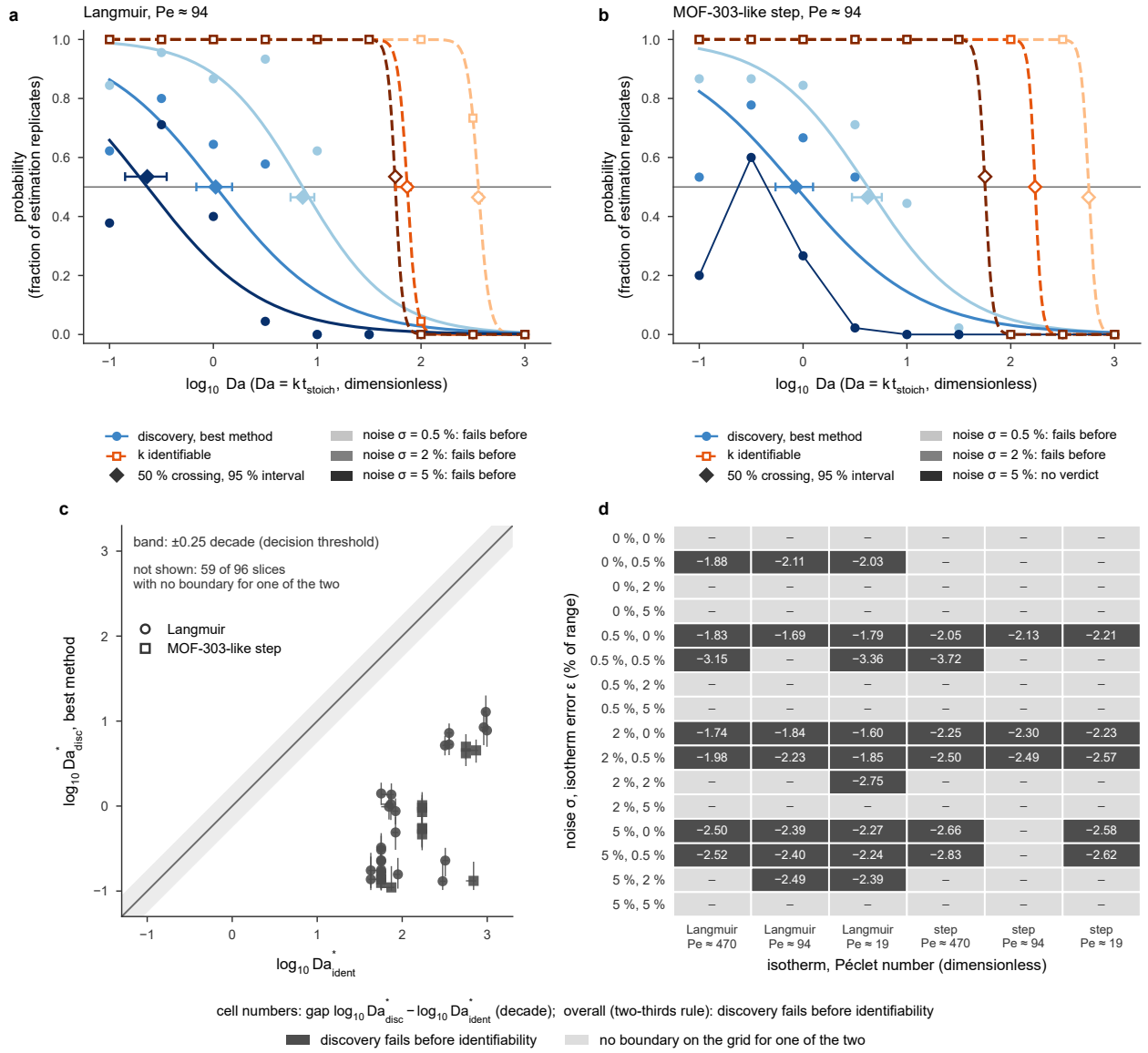


Figure 3: **Discovery fails long before identifiability.** **a**,**b**, Fraction of the 45 estimation replicates in which the best discovery method recovered the LDF law (filled circles, blue, solid) and in which k was practically identifiable (open squares, vermilion, dashed), against Da, for an exact isotherm and noise of 0.5, 2 and 5 % (light to dark), at Pe ≈ 94; **a**, Langmuir sorbent; **b**, MOF-303-like step isotherm. Curves are the logistic fits that define the boundaries, drawn only where a 50 % crossing exists; otherwise the measured fractions are joined. Diamonds mark the crossings, offset slightly from 0.5 so that they do not overlap, with 95 % bootstrap intervals. **c**, $\mathrm{Da}^*_{\mathrm{disc}}$ against $\mathrm{Da}^*_{\mathrm{ident}}$ for the 37 slices with both boundaries, with 95 % bootstrap intervals (vertical, discovery; horizontal, identifiability, zero width where the boundary lies between two levels); the grey band is the pre-registered decision threshold (±0.25 decade) around equality. **d**, The verdict of every slice (rows: noise and isotherm error; columns: isotherm and Péclet number) with the gap $\log_{10}\mathrm{Da}^*_{\mathrm{disc}} - \log_{10}\mathrm{Da}^*_{\mathrm{ident}}$ in decades, negative when discovery fails first. Light grey cells have no boundary on the grid for one of the two.

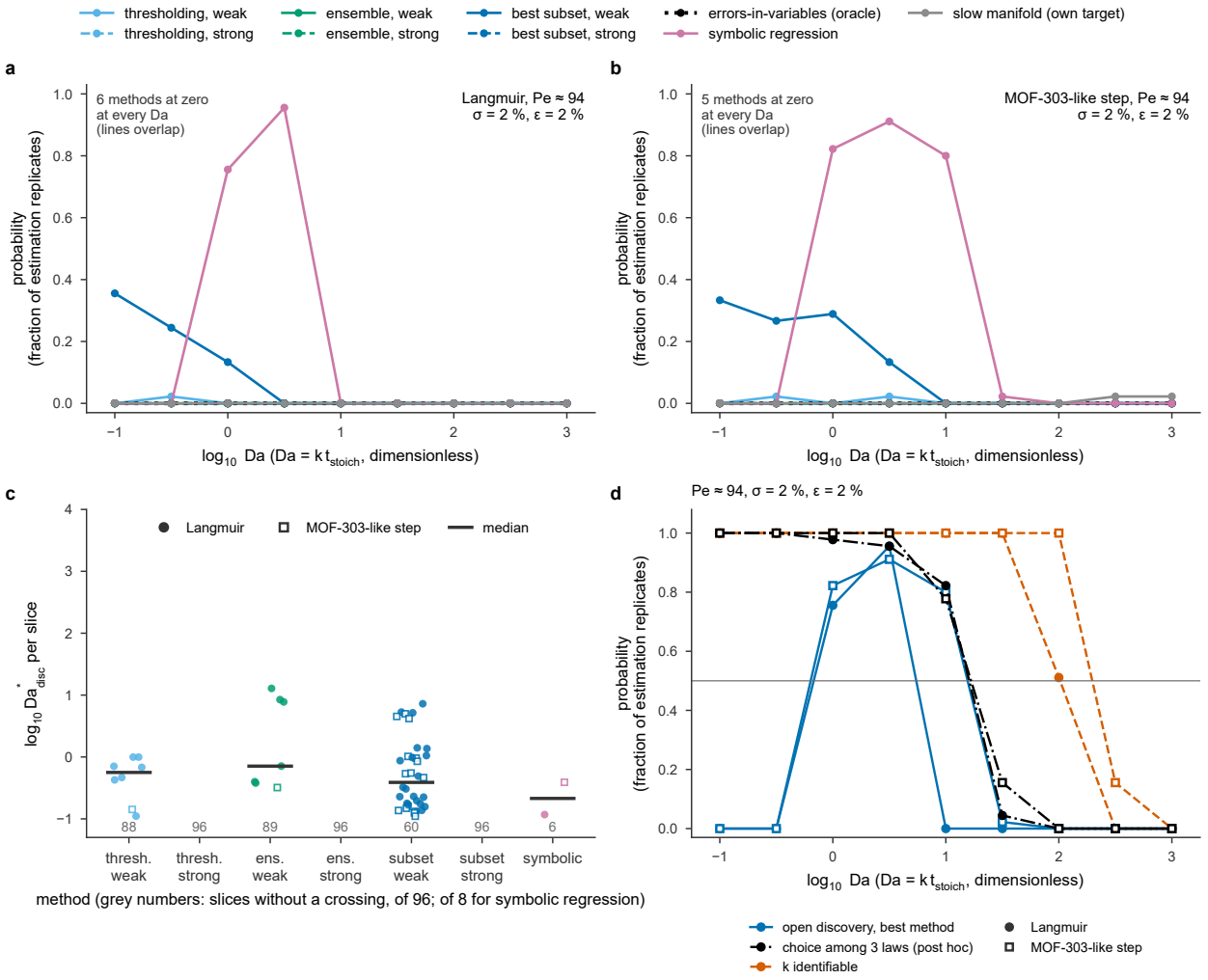


Figure 4: **Methods compared.** **a**,**b**, Fraction of the 45 estimation replicates in which each method recovered the law, against Da, at 2 % noise and 2 % isotherm error, Pe ≈ 94; **a**, Langmuir; **b**, step isotherm; symbolic regression in purple; solid lines, weak form; dashed, strong form. **c**, $\log_{10} \mathrm{Da}^{*}_{\mathrm{disc}}$ of each method in every slice where it crosses 50 % (filled circles, Langmuir; open squares, step; bar, median); grey numbers count the slices without a crossing, of 96 (of 8 for symbolic regression, run on a subgrid). **d**, Fraction of the 45 estimation replicates in which the LDF law was chosen among three candidate laws by likelihood (black, dash-dotted; post hoc), recovered by the best discovery method (blue) or had an identifiable k (vermilion, dashed), at Pe ≈ 94, 2 % noise and 2 % isotherm error; circles, Langmuir; open squares, step isotherm.

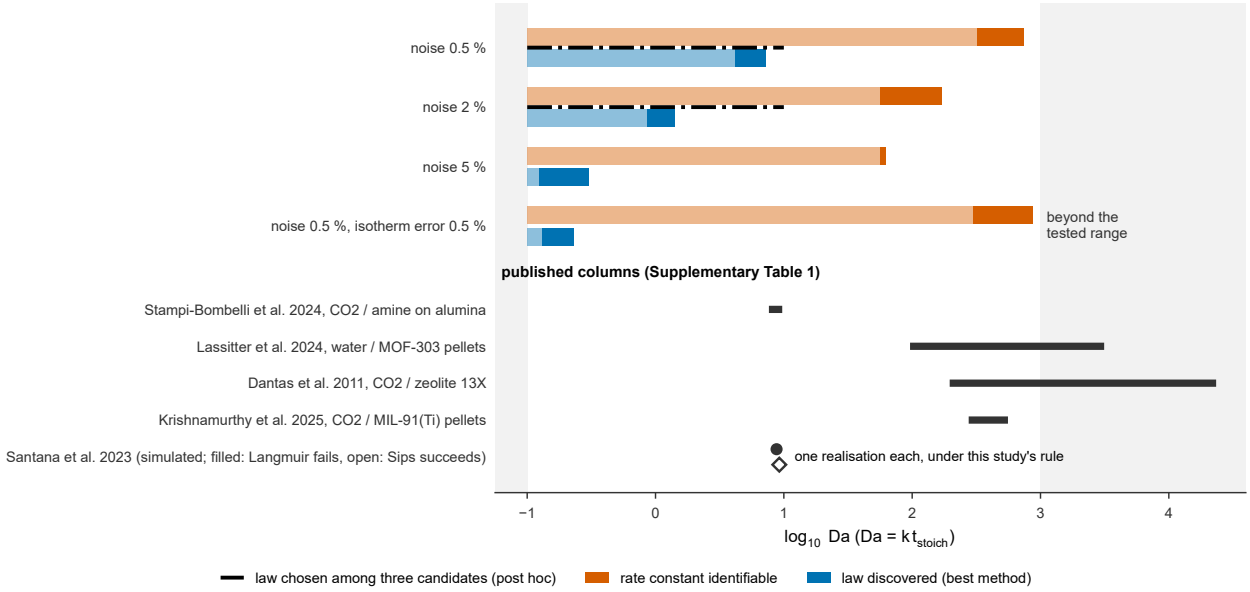


Figure 5: **What column data allow, by Damköhler number.** For each condition (probes with uptake, O1), bars run from the bottom of the grid (Da = 0.1) to the 50 % boundary, and the darker end spans that boundary over the isotherms and Péclet numbers: discovery by the best method (blue), practical identifiability of k (vermilion), and choice of the LDF law among three candidate laws in every replicate (black, dash-dotted; post hoc, Pe $\approx$ 94, exact isotherm); all with 200 observation times per run (denser sampling raises discovery most, Results). Below, the Da ranges of published packed beds (Supplementary Table 1) and the two simulated columns of the closest prior study [12], computed from its published equations, with the outcome of its readout under this study's success rule. Shaded, outside the tested range.

Table 1: **Pre-registered hypotheses and their verdicts.** Words in quotation marks are the pre-registered verdict words, except the two "no verdict" entries, which are the analysis code's words where no pre-registered rule applies. The gap is $\log_{10} \mathrm{Da}^*_{\mathrm{disc}} - \log_{10} \mathrm{Da}^*_{\mathrm{ident}}$, negative when discovery fails first. Intervals are 95 % bootstrap intervals over the estimation replicates.

	Question	Result	Verdict
Positive control	Do discovery and identifiability both succeed at the lowest Da, without noise or isotherm error ($\geq 90\,\%$)?	Langmuir 135/135 and 135/135; step 135/135 and 135/135	Both isotherms enter every verdict
H2b	Does discovery fail before, with, or after identifiability (± 0.25 decade)?	37 of 96 slices with both boundaries; gap -1.60 to -3.72 decades (median -2.23 Langmuir, -2.50 step)	"*discovery fails before identifiability*" (37 of 37 slices)
H2b, k-aware	As H2b, with success also requiring $\hat{k}$ within a factor of 2	39 of 96 slices with both boundaries; the 37 verdicts unchanged	"*discovery fails before identifiability*" (39 of 39)
H2b, any method (post hoc)	How small a gap does any method allow?	Each slice's best method judged on the estimation replicates gives the same 37 gaps and two more ($\sigma = \varepsilon = 0.5\,\%$; -3.22 and -3.73), all "*fails before*"; symbolic regression (four slices, success in a window of Da) last reached 50 % 1.0 to 1.4 decades below $\mathrm{Da}^*_{\mathrm{ident}}$	Descriptive
H2a	Is the boundary at one value of $R = \delta/\varepsilon_{\mathrm{eff}}$ (spread < 0.5 decade)?	One of 12 conditions crosses ($R^* = 123$, 113 to 135)	"*no verdict: fewer than two conditions have a crossing*"
H2c	Does errors-in-variables fitting move the boundary up by ≥ 0.25 decade (two thirds of slices) at $\varepsilon > 0$?	No slice with a boundary for both estimators	"*no verdict*" at every ε
H2d (exploratory)	Is a loading-dependent rate discovered for the step isotherm?	At $\mathrm{Da} = 1$ in 77 to 100 % of replicates (best method); at most 58 % elsewhere	Descriptive

closed-form solutions. Uptake followed one of two laws,

$$\text{L1:}\quad \frac{\partial q}{\partial t} = k\,(q^*(c,T) - q), \tag{1}$$

$$\text{L2:}\quad \frac{\partial q}{\partial t} = k(q)\,(q^*(c,T) - q), \qquad k(q) = k_0\Big[1 - 0.8\,s\Big(\frac{q/q_{\max} - \theta}{0.05}\Big)\Big], \tag{2}$$

where s is the logistic function and θ is the fractional loading at the isotherm's step midpoint, so that under L2 the rate falls to 20 % of k_0 across the step. Two adsorbents were simulated with the same column geometry and flow (Supplementary Table 2): the companion study's favourable Langmuir sorbent, and water on a MOF-303-like framework, a cooperative (Type V) isotherm with its step at $P/P_0 = 0.15$, fed at 30 % relative humidity and 30 °C. The Damköhler number $\mathrm{Da} = k\,t_{\mathrm{stoich}}$ counts kinetic time constants per passage of the adsorption front, with t_{stoich} the stoichiometric time of the column (7200 s for Langmuir and 31 050 s for the step isotherm). Nine levels from 10^{-1} to 10^{3} at half-decade spacing therefore span k from 1.39×10^{-5} to $0.139\,\mathrm{s}^{-1}$ for Langmuir and from 3.22×10^{-6} to $0.0322\,\mathrm{s}^{-1}$ for the step isotherm. Three axial-dispersion coefficients, $D_L \times 0.2$, $\times 1$ and $\times 5$ of the default, completed the grid. They give column Péclet numbers $\mathrm{Pe} = uL/D_L$ of about 470, 94 and 19, and the levels are named by these values below (the code and the pre-registration label them by the D_L multiplier, so its "×0.2" level is the highest Pe). The grid has 81 solves at 400 cells and 400 stored times, each run until the outlet reached 95 % breakthrough, plus 9 isothermal solves for a control described below. The manifest that records every parameter is public.

Observation models, noise and isotherm error

Every method and every identifiability calculation read the data through one function, so all saw the same corrupted observations. Three observation models were used: interior probes of c, q and T (O1), interior probes of c and T with q hidden (O2), and the outlet $c(t)$ and $T(t)$ only (O3). O1 and O2 used 20 equally spaced probes; all models used 200 times spread evenly over the solved horizon. Noise was independent Gaussian with standard deviation $\sigma \in \{0, 0.5, 2, 5\}$ % of each channel's range. Methods were never given the true isotherm. They were given a measured one, $q^*_{\mathrm{meas}}(c,T) = q^*(c,T)\,[1 + \varepsilon\, g(c/c_{\mathrm{in}})]$ with $\varepsilon \in \{0, 0.5, 2, 5\}$ %, where g is a smooth random function of four Fourier modes on $[0,1]$, scaled to unit root-mean-square and drawn once per replicate. Every random draw was seeded by a hash: the noise by (cell, replicate, σ, observation model) and the isotherm error by (cell, replicate), so each observation can be regenerated on its own and every observation model sees the same measured isotherm. A replicate, one such observation of one cell, was the unit of analysis. Discovery on O1 and O2 and identifiability on O1 used 60 replicates per cell; the costlier O2 and O3 identifiability profiles used 3, and the M1 demonstration 1.

Discovery methods

Each method ran in the strongest configuration we could verify on known-answer systems before the pre-registration was frozen, and a method that could not be run that way was reported as such rather

than replaced by a weaker one. The candidate library was $\{1, c, q, T, cq, q^2, c^2, q^*_{\mathrm{meas}}\}$; under L1 the true law is $k(q^*_{\mathrm{meas}} - q)$. The methods were sparse regression on a Savitzky–Golay estimate of $\mathrm{d}q/\mathrm{d}t$ (strong form) and on its weak, integral form with compactly supported test functions [3, 41], each with three solvers: sequentially thresholded least squares [1], a bagged ensemble with an inclusion threshold of 0.6 [5], and exact best-subset selection by enumeration (M4; exact subset selection as in [6], here scored by the Bayesian information criterion); errors-in-variables best-subset regression given the true error levels (M5, an oracle); implicit sparse regression without any isotherm on an isothermal control (M6, SINDy-PI [42]); symbolic regression (M7, PySR [7]); a universal differential equation with a neural network in the uptake term, trained through the column model and read out by sparse and symbolic regression (M1, the pipeline of [12]); and regression of the state on q-free features, which returns the isotherm where uptake is at equilibrium (M9 [25]). On O2, uptake was first recovered from the gas-phase mass balance, and the three sparse solvers were then run on it in both forms; M5, M7 and M9 were not run on O2, as the pre-registered applicability table states. 2 h per fit on the machine available, so it ran as a declared demonstration: Langmuir, Pe ≈ 94, $\sigma = 2\,\%$, $\varepsilon = 0$, O3, the nine Da levels, one replicate. The best parameters seen during training were kept. As first run, a column solve that exhausted its step budget during training ended the fit with no readout (four of the nine fits); the corrected code keeps the best parameters in that case, and both runs are reported (Supplementary Note 2). A Kolmogorov–Arnold network [43] failed its known-answer test and its cost gate before the freeze and was not run on the grid. Method settings, solver thresholds and their known-answer tests are listed in Supplementary Note 3.

Success

Discovery succeeded when the selected terms were exactly those of the true law, every sign was correct and, for L1, the ratio of the q^*_{meas} and q coefficients was -1 within 10 %. The value of k itself was not required. For L2 (H2d), the library holds no term for a rate falling sigmoidally across the step, so the rule asks instead whether q^*_{meas} and q are selected and the rate implied by the selected terms, $-\partial f/\partial q$ averaged in ten bins of loading, decreases across the bins (Spearman $\rho < -0.8$) to at most 60 % of its value in the lowest bin. A method that raised an error counted as a failure; no replicate, cell or output was excluded.

Identifiability

Practical identifiability of k was judged by profile likelihood [36] with the true law known, on the same observations and with the same measured isotherm as the discovery methods: k was practically identifiable when the 95 % profile interval (likelihood-ratio threshold 3.84) lay within $[k/2, 2k]$. On O1 the likelihood summed over all probes the fit of the known law, integrated exactly for piecewise-linear $q^*(t)$ along the observed c and T. On O2 and O3 every point of the profile was a full column solve. The O1 likelihood involves only the local uptake law along the observed c and T, so errors in dispersion, heat transfer or geometry do not enter it; on O2 and O3 the column model is taken as exact, as it is in the simulations. The profile searched a coarse logarithmic grid of

k and refined it around the interval (41 points per pass on O1, 21 on O2 and O3, as pre-registered). Three corrections to the identifiability profiles were made during the study and are reported in full (Supplementary Note 2). First, on O2 and O3 a measured isotherm with $\varepsilon > 0$ could stall the stiff solver at large trial k for days; each solve was therefore capped at 600 s and a capped solve was scored as a failed solve (amendment A5; two O3 rows had capped solves, none next to its interval). Second, the O1 grid first spanned 10^{-5} to $10\,\mathrm{s}^{-1}$, which excluded the step cells at Da = 0.1; it was widened to 10^{-7} to $10\,\mathrm{s}^{-1}$, a peak on the grid edge was made an error, and all O1 rows were recomputed (amendment A6). Before that recomputation, the O1 refinement was found to use 21 points per pass instead of the pre-registered 41 and was set to 41, so every corrected O1 row uses the declared settings. Third, a later review of the code found that the O1 update, meant to be exact, carried a wrong coefficient: it differed from the exact solution only by a term of order $(kh)^2$ (h the interval between observations), so a check made at small kh had passed, but it misfit the data when kh was of order one or larger, that is, at high Da. The update was corrected and tested against a numerical ODE solver for kh from 10^{-4} to 100, and all 51 840 O1 rows were recomputed with the analysis unchanged (amendment A7 records this correction and the refinement setting).

Boundaries and verdicts

For each slice the boundary Da^* was the 50 % crossing of an unpenalised logistic regression of replicate outcomes on $\log_{10}\mathrm{Da}$, with a percentile bootstrap interval (2000 resamples within each Da level). No boundary was reported, and no verdict issued, when the fitted curve did not cross 50 % inside the tested range or rose with Da instead of falling; the range was reported instead. The boundary is the crossing of the fitted curve, so a slice whose observed success stays just below 50 % can still have one: in one slice (Langmuir, Pe $\approx$ 19, 2 % noise and isotherm error) success was 0.49 at the two lowest levels and the curve crosses 50 % at Da = 0.16. In all, 10 of the 37 discovery boundaries lie within 0.25 decade of the lowest level, Da = 0.1; their slices have gaps of 2.3 to 3.7 decades, so the narrowest gaps do not depend on them. Before the freeze the estimator's coverage on planted boundaries was 0.94 for a nominal 0.95, with a bias of +0.012 decade. The best discovery method of each slice was chosen on replicates 0–14 and its boundary estimated on replicates 15–59, which removes the inflation that choosing and estimating on the same replicates causes (0.07 decade in simulation); the identifiability boundary was estimated on the same 45 replicates. The best method was the one with the largest crossing on the selection replicates, or, if none crossed, the one with the highest mean success. Only rate-law methods that ran on every cell of the slice competed: M5, which knows the true error levels, and M9, which targets the isotherm, were excluded.

Three hypotheses were confirmatory, each with verdict words fixed in advance. H2a: success collapses onto one curve in $R = \delta/\varepsilon_{\mathrm{eff}}$, the driving force relative to the combined error, with the spread of R^* across conditions under half a decade. H2b: on O1, for each slice (σ, ε, isotherm, Pe; 96 in all), the gap $\log_{10}\mathrm{Da}^*_{\mathrm{disc}} - \log_{10}\mathrm{Da}^*_{\mathrm{ident}}$ was called "*discovery fails before identifiability*" at -0.25 decade or less, "*discovery outlives identifiability*" at +0.25 or more, and "*discovery fails with*

identifiability" in between; the hypothesis-level verdict was the word held by at least two thirds of the slices with a verdict. Slices share ground-truth solves and isotherm-error draws, so they are not independent, and the two-thirds rule summarises them without a formal error rate. H2c: errors-in-variables regression moves the discovery boundary upwards by at least 0.25 decade in at least two thirds of the slices with a verdict at a given $\varepsilon > 0$, tested at $\sigma = 2\,\%$ on O1 in the strong form against the best strong-form method that is not errors-in-variables, in 18 slices (three ε, two isotherms, three Péclet numbers); a slice has a verdict only if both estimators have a boundary on the grid, and the refined grid placed around the comparator's boundary is solved only if that boundary exists. With 45 estimation replicates the H2b rule returned a wrong directional verdict with probability 0.001 when there was no gap, and the correct one with probability 0.965 at a gap of 0.40 decade, the smallest effect of interest. A positive control guarded every verdict: in each isotherm's easiest cells (lowest Da, $\sigma = \varepsilon = 0$, O1, the three Péclet numbers pooled, where the pre-registration names one easiest cell) the best method had to succeed and k had to be identifiable in at least 90 % of the estimation replicates, or that isotherm's verdicts were withheld. O2 and O3 were reported descriptively, and H2d, whether the loading-dependent rate of L2 can be discovered, was exploratory.

Pre-registration and amendments

The design was frozen on 11 October 2026 (commit `7dcc2fe`) before any discovery method ran on any grid cell, and every confirmatory driver refused to run unless the pre-registration matched its latest freeze commit. GitHub's server-side event log records the freeze commit reaching the public repository on 11 October 2026 at 08:36 UTC, 30 hours before the grid started (a copy of the log is archived with the results). (The study's history was moved from the companion project's repository into its own; the pre-registration's amendment headers cite the original commit identifiers, and a published table maps them to the ones given here.) Seven dated amendments are appended to it. A1 to A4 were made before the grid ran: 60 replicates split into selection and estimation sets, the smallest effect of interest, the oracle kept out of the pooled verdicts, Pe as a slicing axis, errors counted as failures, the decision to run the grid before peer review, and the M1 demonstration. A5 (the solve cap) was made while the grid ran, and A6 (the O1 grid range) and A7 (two corrections to the O1 code, recorded without changing the declared method) after the verdicts were first computed. Five errors in our own code and one solver setting were found and corrected after the grid started. The one that mattered most, a wrong coefficient in the O1 likelihood found by an independent review, moved 36 of the 71 identifiability boundaries present in both runs to lower Da (none higher), brought 22 more onto the grid and raised the number of slices with a verdict from 30 to 37. The verdicts as first run, before the last correction and corrected are all public, and the paper reports them (Extended Data Fig. 8, Supplementary Note 2); no correction changed a pre-registered verdict word. Every deviation of the code from the pre-registration, and every pre-registered analysis that was run only after the verdicts or not run as declared, is listed in Supplementary Note 3.

Exploratory analyses

Every exploratory analysis was declared in writing, with what it would compute, before it ran, and none enters a confirmatory test; the descriptive checks listed at the end of this section were not declared in advance. Three were declared before the verdicts: M1 trained at the scale a universal differential equation needs (E1), the Kolmogorov–Arnold network on its declared subgrid (E2), and how often an interval that meets $[k/2, 2k]$ misses the true k (E3). E1 needs a graphics processor and has not run; it is not reported here. Eight more were declared after the verdicts, prompted by a review of the draft, and are labelled post hoc: H2b with matched success criteria for the two tasks (E4), the conditioning of the library against Da (E5), choice among three candidate laws by likelihood (E6), a systematic shift of the step of the measured isotherm (E7), the sampling rate, 200 against 400 observation times (E8, declared after E6's results), E6 with q^*_{meas} computed from noise-free c and T (E9, declared after a second review), E6 on noise-free data sampled up to 20 times more finely (E10, declared after E9), and every boundary at about 2,000 observation times per run, with the fields re-solved at finer time resolution (E11, declared after a third review, with its reading fixed in advance). E6 fits each law by implicit Euler along the same upsampled path of q^*_{meas} as the identifiability profile, with its rate constant profiled on a logarithmic grid and refined by a bounded one-dimensional search; all three laws have one parameter, so the Bayesian information criterion picks the largest likelihood. In E8 every method runs unchanged, so its windows, defined in samples, span half the time at 400 samples. Descriptive checks made after the verdicts are labelled post hoc wherever they are used: the accuracy of the O1 profile with exact data, H2b without the two highest Da levels, a classification of the slices without a verdict, each slice's best method judged on the estimation replicates, the last level at which symbolic regression reached 50 %, the O3 identifiability boundaries against the last level of O1 discovery, the boundaries in units of kh, and the number of terms selected against noise. All are reported in Supplementary Note 1.

Boundaries with perfect separation

Where every replicate is identifiable at one level of Da and none at the next, the unpenalised logistic fit has no finite maximum-likelihood slope: the optimiser stops at a steep curve whose crossing lies between the two levels, near their geometric midpoint. Every bootstrap resample reproduces the same data, so the interval has zero width. Such a boundary is reported as lying between those two levels; the gap used by H2b is computed from it exactly as from any other crossing. Such a boundary is interval-censored. This was so for the identifiability boundary in 18 of the 37 slices with a verdict; placing each of these at the lower of its two levels, nearest the discovery boundary, every gap is still at least 1.35 decades, so no verdict depends on where in its interval the boundary lies, and the gaps quoted in the main text are point estimates within these bounds.

Statistics and reproducibility

The unit of analysis was a replicate, one independently seeded observation of one simulated cell. Each cell had 60 replicates for O1 and O2 discovery and O1 identifiability: replicates 0–14 selected each slice's best method and replicates 15–59 estimated the boundaries. Boundaries are 50 % crossings of unpenalised logistic fits with 95 % percentile bootstrap intervals (2,000 resamples within levels). No replicate, cell or output was excluded; errors counted as failures. All random draws are seeded by hashes of their keys, so any observation can be regenerated alone, and the design, thresholds and verdict words were fixed before the grid ran.

Software and computation

The grid ran on a single computer without a graphics processor, one job at a time, from 12 October to 1 November 2026; the corrected O1 identifiability rows and the four M1 refits were computed from 1 to 4 November. After the last correction, the O1 identifiability rows were recomputed on 8 November on cloud CPU sessions (Kaggle) with the same code, and a random sample of rows recomputed locally agreed to a relative 10^{-6}; the exploratory model-selection and sampling analyses ran the same way. Package versions are pinned in the repository.

Use of AI tools

During the preparation of this work the author used a generative AI assistant (Claude, Anthropic, used through the Claude Code and Claude desktop applications from September to November 2026, most recently the Claude Opus 5.5 model) as a tool to assist with drafting and language editing, with writing and checking code, and with checking bibliographic records. All code is covered by unit and known-answer tests, the build script checks the numbers in the text against the results files, and the bibliographic records were checked against Crossref or the publisher. The author designed the study, made every scientific decision, reviewed and edited all content, and takes full responsibility for the content of the published article.

Data availability

The ground-truth manifest, every results file, the run log and the as-run and corrected verdicts are public in the study repository and archived on Zenodo [44]. The ground-truth fields are regenerated from the manifest by `p2_generate.py`. Source data for Figs. 2–5 are provided with this paper.

Code availability

All code is public at `https://github.com/abdulhamidbatayhi123/ai-kinetics-discoverability` under the MIT licence and archived with the data [44]. The solver is imported unchanged from the companion study's archive [45].

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Author contributions

A.B. is the sole author and conceived and designed the study. The use of AI tools is described in Methods.

Competing interests

The author declares no competing interests.

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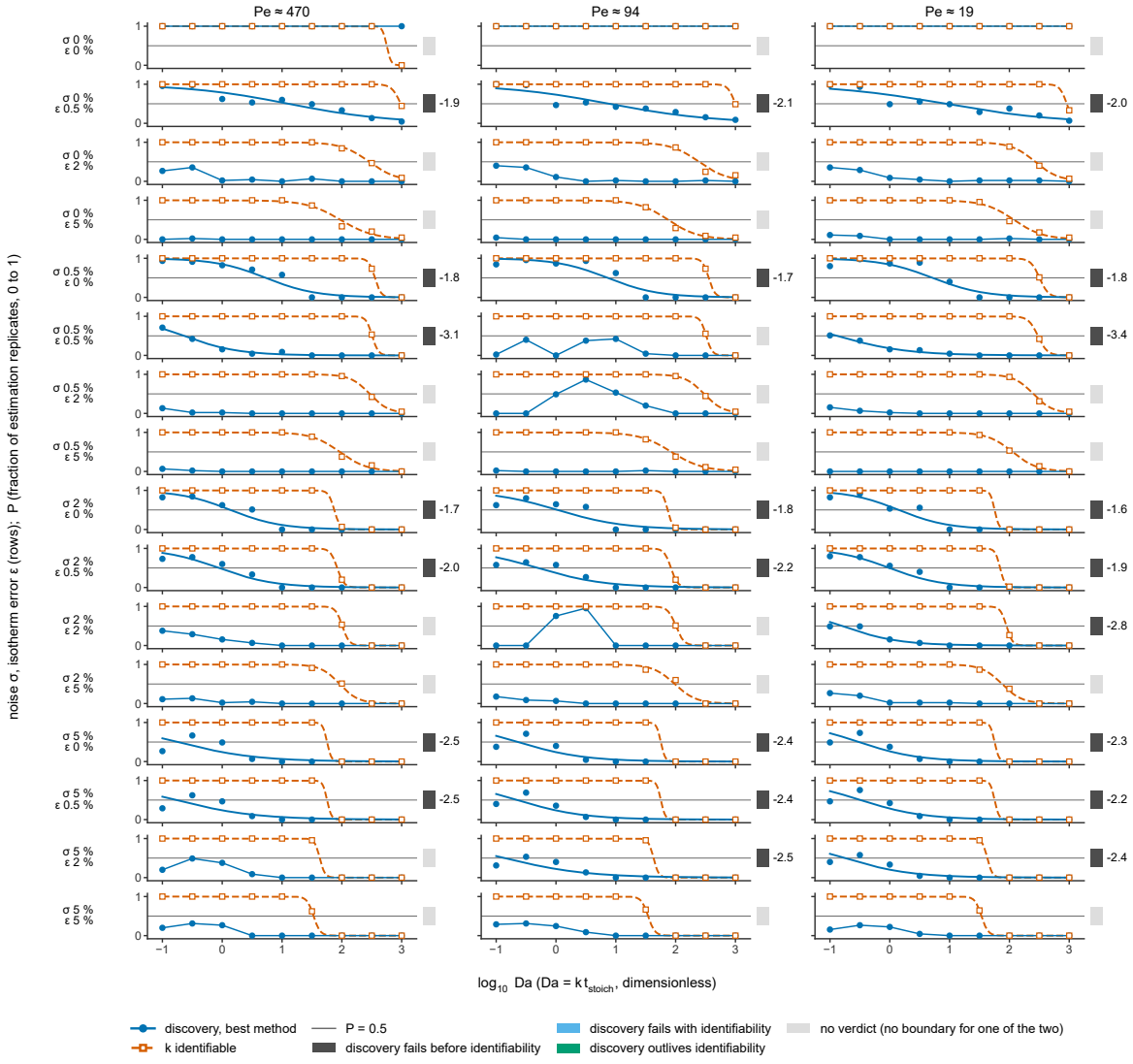
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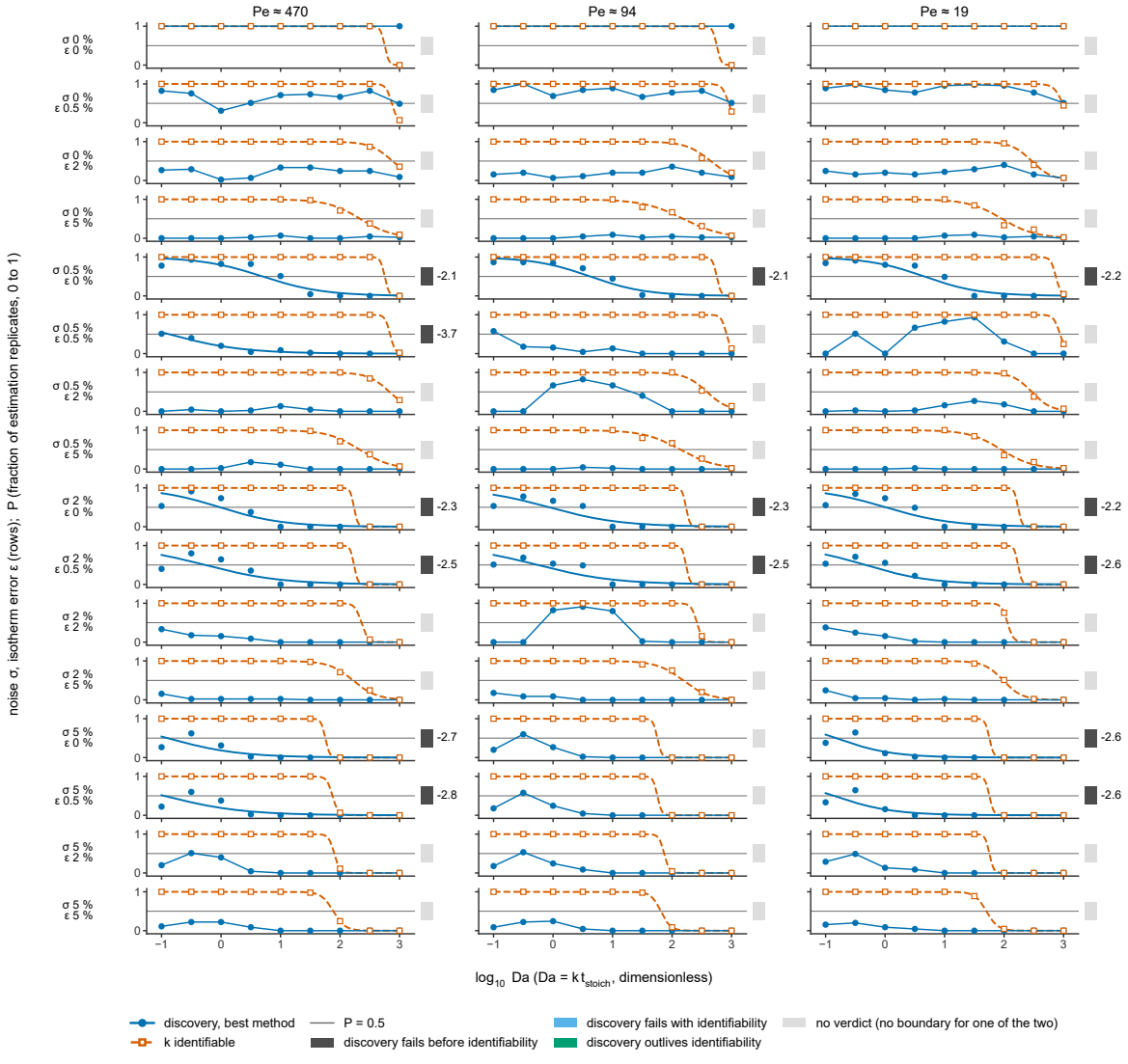
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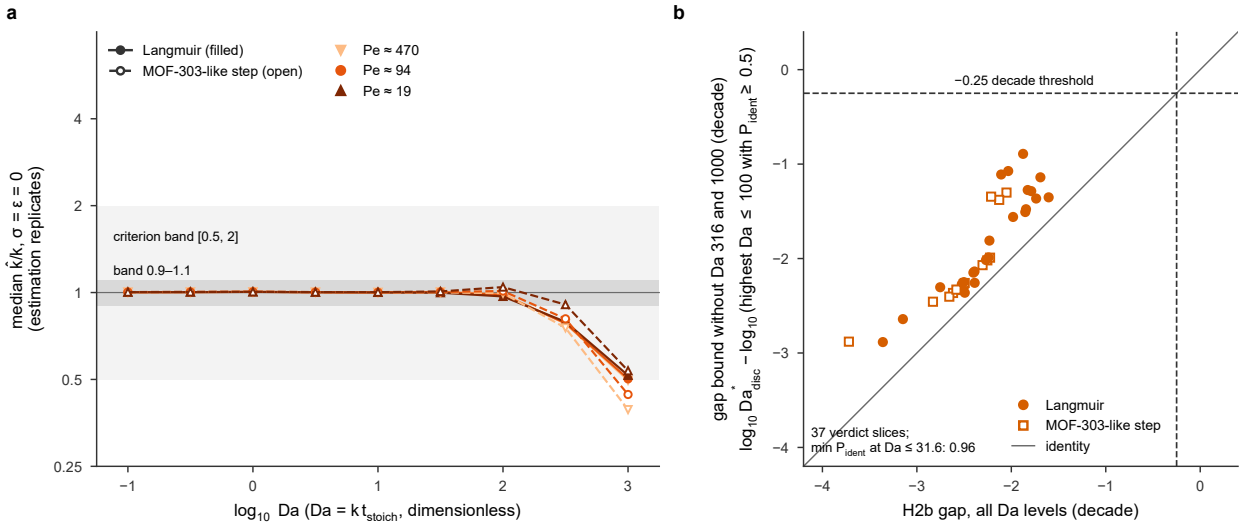
Extended Data



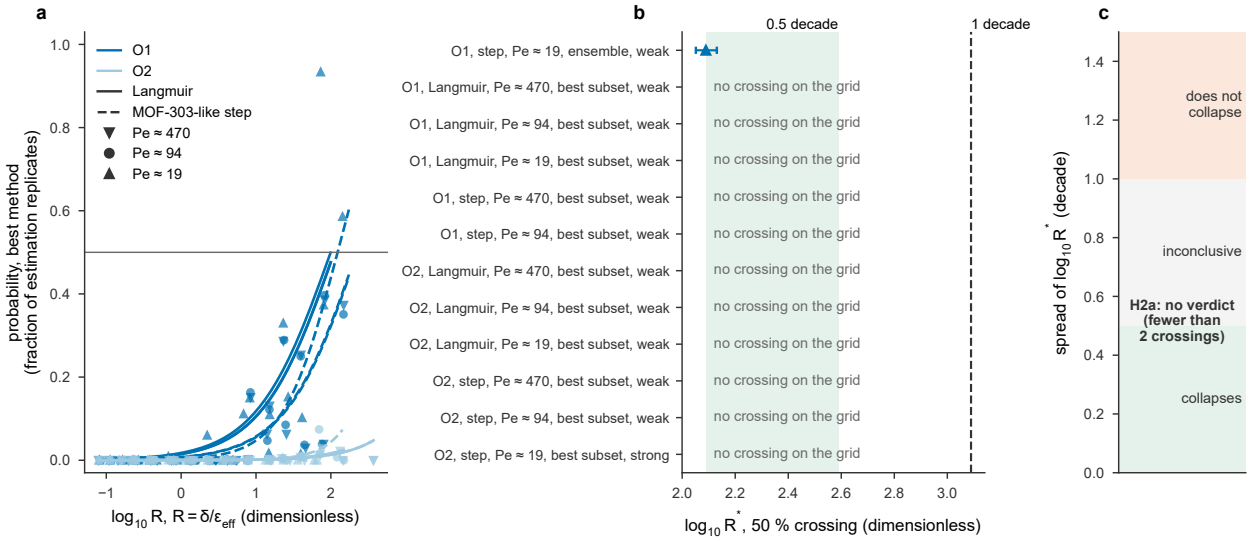
Extended Data Fig. 1: **Every slice of the confirmatory test, Langmuir sorbent.** Fraction of the 45 estimation replicates in which the best method recovered the law (blue circles) and in which k was identifiable (vermilion open squares), against $\log_{10}$ Da; rows, noise σ and isotherm error ε; columns, Péclet number. Smooth curves are the logistic fits that define the boundaries, drawn only where a 50 % crossing exists; otherwise the measured fractions are joined. The square beside each panel gives the verdict (dark grey, "*discovery fails before identifiability*"; light grey, no boundary on the grid for one of the two) and the number gives the gap in decades.



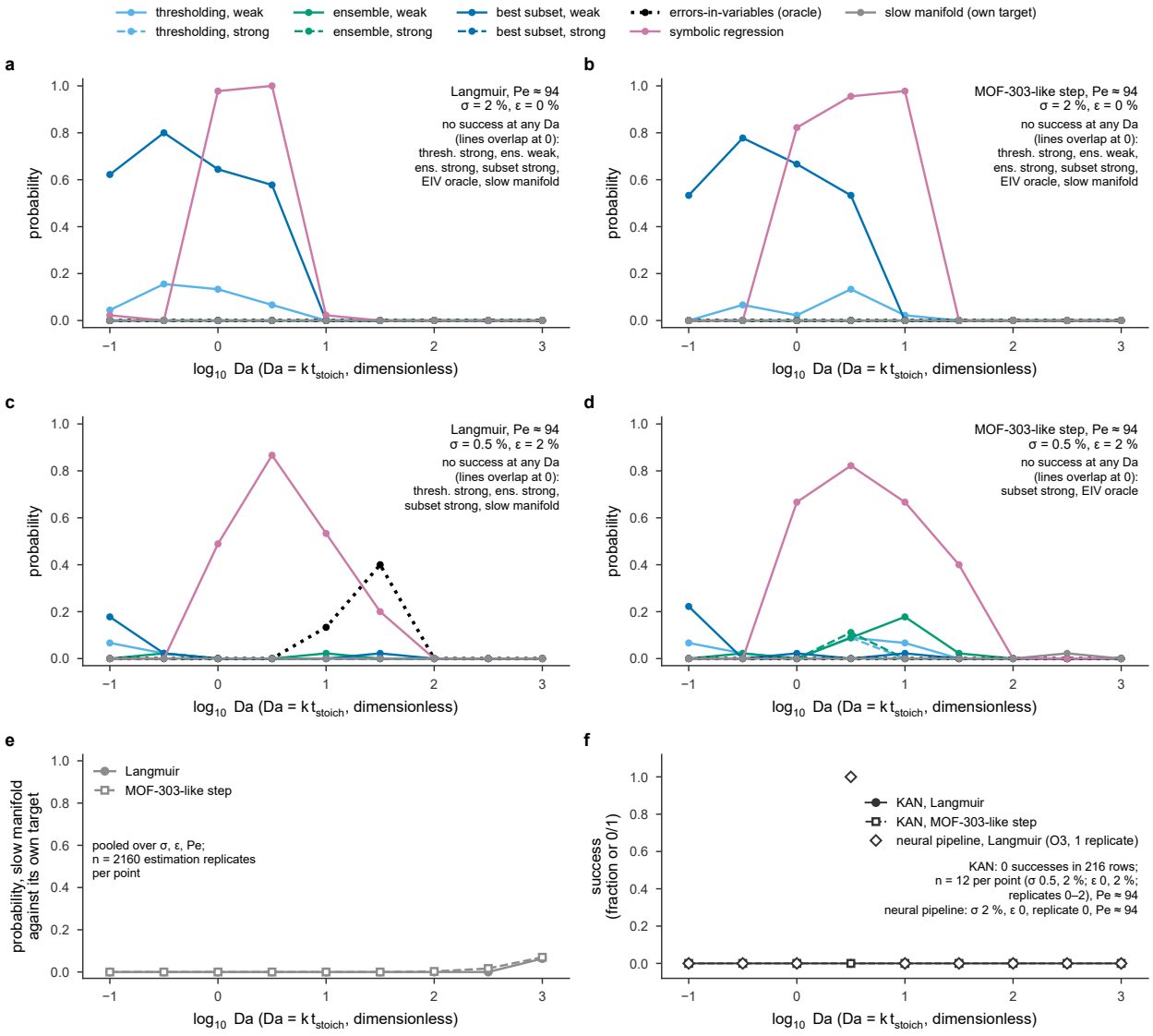
Extended Data Fig. 2: **Every slice of the confirmatory test, MOF-303-like step isotherm.** As Extended Data Fig. 1.



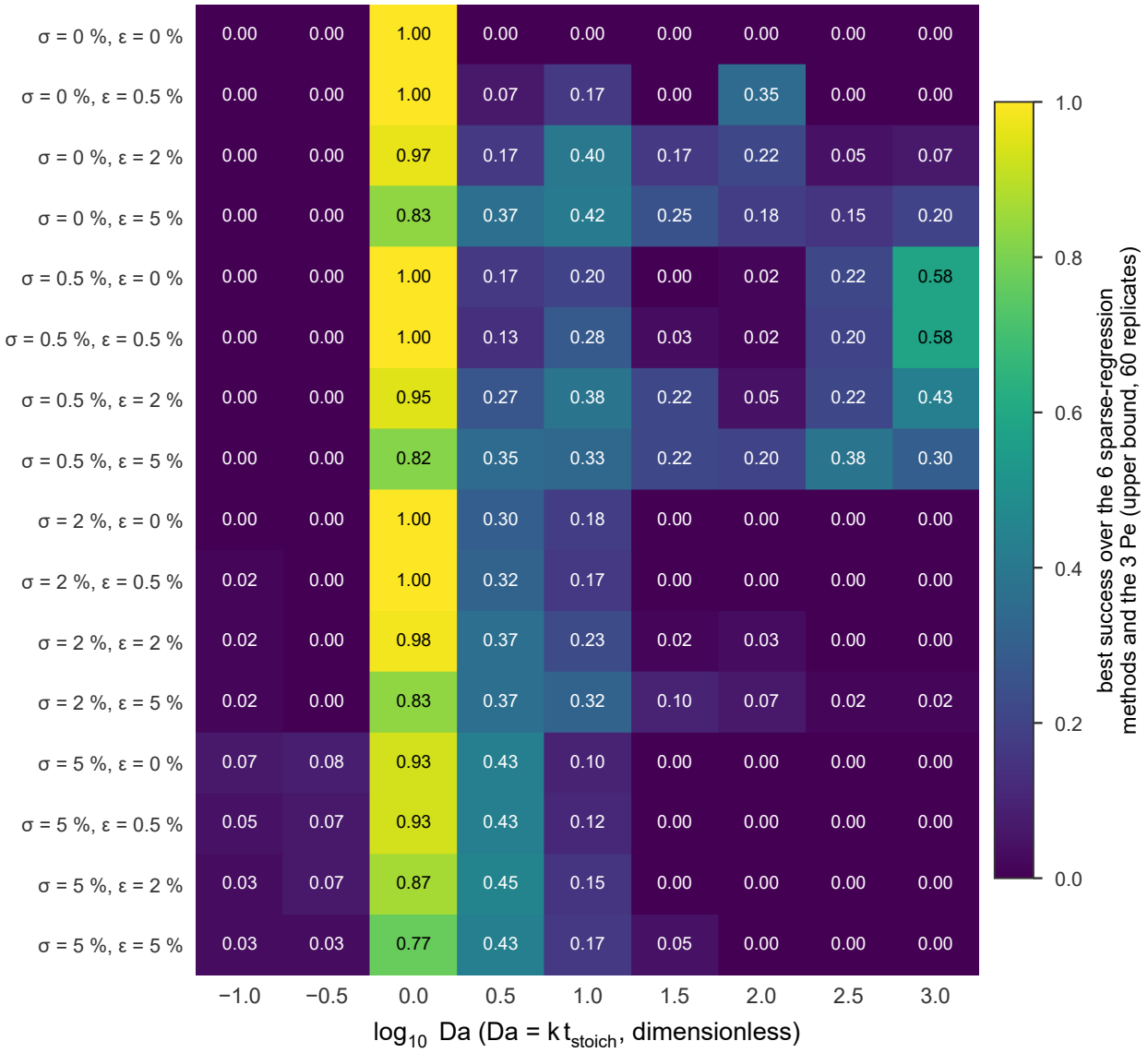
Extended Data Fig. 3: **Accuracy of the O1 identifiability estimator, and the confirmatory test without the top two levels (post hoc).** **a**, Median ratio of the estimated to the true k with exact data ($\sigma = \varepsilon = 0$), against Da, for both isotherms and three Péclet numbers; dark band, 0.9 to 1.1; light band, the criterion's $[0.5, 2]$. **b**, For the 37 slices with a verdict, the gap against the gap computed from the highest level up to Da = 100 at which k was identifiable in half the replicates (so without Da = 316 and 1000); dashed lines, the −0.25 decade threshold.



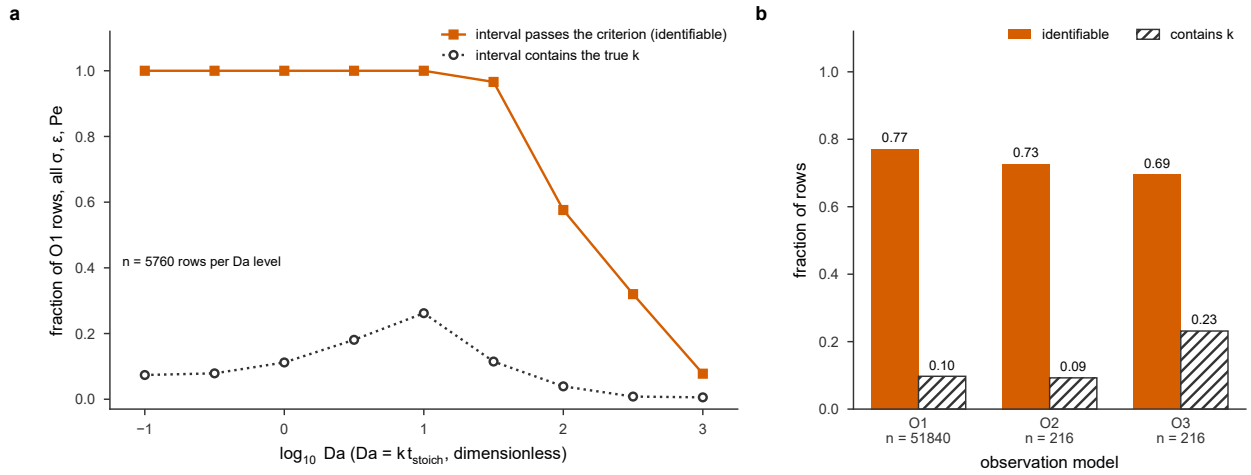
Extended Data Fig. 4: **The error-ratio test (H2a) could not be decided.** **a**, Success of the best method in each condition against $R = \delta/\varepsilon_{\text{eff}}$, the realised driving force over the combined observation and isotherm error, on probes with uptake (O1, dark blue) and without (O2, light blue); points, binned fractions; curves, logistic fits. **b**, The 50 % crossing R^* of each condition with its 95 % interval; one of the twelve conditions crosses. **c**, The pre-registered decision regions for the spread of $\log_{10} R^*$; with fewer than two crossings, the test has no verdict.



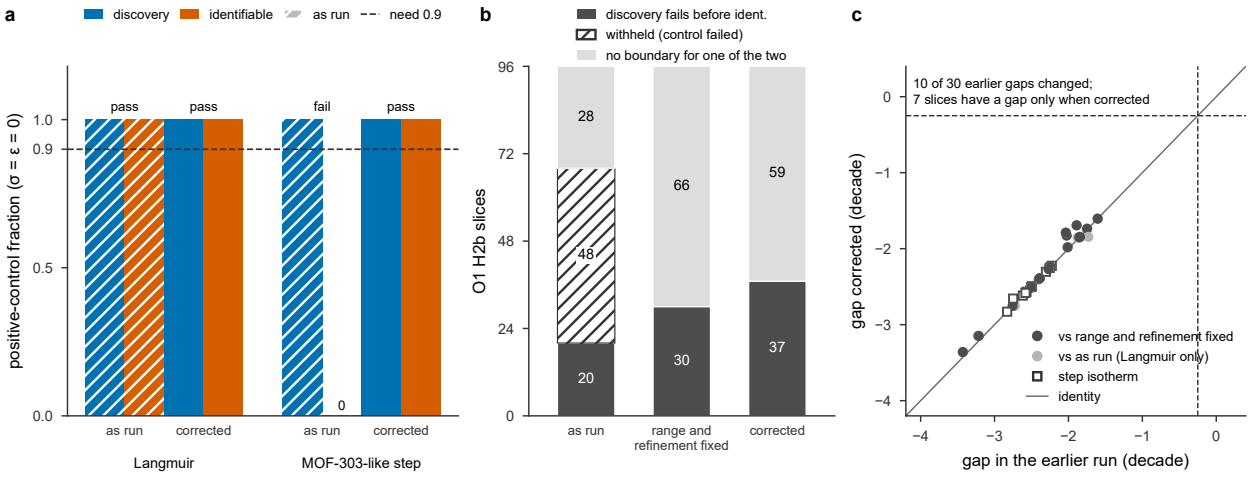
Extended Data Fig. 5: **Every method against** Da**, and the methods outside the confirmatory pool. a**–**d**, Fraction of the 45 estimation replicates in which each method recovered the law, at Pe ≈ 94, with 2 % noise and an exact isotherm (**a**, Langmuir; **b**, step) and with 0.5 % noise and 2 % isotherm error (**c**, **d**); each panel lists the methods with no success at any Da. **e**, Slow-manifold discovery against its own target, pooled over noise, isotherm error and Péclet number. **f**, The Kolmogorov–Arnold network (KAN; exploratory, declared before the verdicts; 0 of 216 fits) and the neural-network demonstration (one replicate per level).



Extended Data Fig. 6: **Loading-dependent rate (H2d, exploratory).** Best success over the six sparse methods and three Péclet numbers, an upper bound for any single method, for each noise and isotherm-error pair (rows) and Da (columns), over all 60 replicates; values printed in the cells.



Extended Data Fig. 7: **Precision against coverage of the identifiability criterion (exploratory, declared before the verdicts).** Fraction of identifiability rows that meet the criterion (95 % interval inside $[k/2, 2k]$) and fraction whose interval contains the true k, by Da on O1 and in total on O2 and O3.



Extended Data Fig. 8: **Results as first run, after intermediate corrections, and corrected.** **a**, Positive-control fractions as first run (hatched) and corrected; the step isotherm failed as first run because of the identifiability search range. **b**, Slices by verdict as first run, after the search range and refinement were corrected, and after the final correction of the O1 update (the primary result). **c**, Corrected gap of each slice against its gap in each earlier run (dark grey, after the range and refinement corrections; light grey, as first run, Langmuir only, the step slices having been withheld). No correction changed a verdict word (Supplementary Note 2).